

ECG Arrhythmia Classification

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Abstract—Electrocardiogram (ECG) interpretation is a critical tool for detecting cardiac abnormalities and assisting clinical decision-making. This document addresses the arrhythmia classification problem by developing and optimizing a machine learning model. The ECG dataset that was used contains 32 different numerical interpretations of ECG, source ECG recording identifier, and 5 different heartbeat classes. However, the Unknown Beat (Q) was excluded from the final modeling dataset due to its extremely limited number of observations. After establishing the initial classification model using logistic regression, XGBoost was subsequently evaluated together with random oversampling and the Synthetic Minority Oversampling Technique (SMOTE) to address class imbalance. Model selection was primarily based on macro-F1 because of the highly imbalanced class distribution. The final SMOTE-XGBoost model achieved the best results of all, improving balanced multiclass performance compared with the baseline while still exhibiting substantial overfitting.

Keywords—machine learning, logistic regression, softmax, cross entropy, electrocardiogram (ECG), Exploratory Data Analysis (EDA), Principal Component Analysis (PCA), XG Boost, Oversampling, Synthetic Minority Oversampling Technique (SMOTE)

I. INTRODUCTION

Cardiovascular diseases remain one of the important health concerns, and abnormalities in cardiac rhythm, known as arrhythmias, can be found through the analysis of electrocardiogram (ECG) signals. ECG recordings contain information about the electrical activity of the heart and can show changes in cardiac rhythm and morphology. However, manually analyzing large quantities of ECG recordings can be time-consuming and the development of computational methods to identify different heartbeat patterns can be useful.

Machine learning provides a time-efficient analysis by using ECG-derived features to automatically classify heartbeats according to their characteristics. In this project, a multiclass classification model is developed to distinguish between four different heartbeat categories using features extracted from ECG recordings. To establish a working classification model, various methods were used, including data preparation, exploratory data analysis, dimensionality reduction, model implementation, and evaluation. After evaluating the initial model, different ways of solving class imbalance problems are introduced; oversampling and SMOTE. Using those measures, XGBoost is used as an improvement of the initial model.

II. OBJECTIVE

The objective of this project is to develop a multiclass machine-learning model to classify ECG heartbeats into five categories: Normal (N), Supraventricular Ectopic Beat

(SVEB), Ventricular Ectopic Beat (VEB), Fusion Beat (F), and Unknown Beat (Q). It is important to mention that even though the initial dataset contains the data for Unknown Beat (Q), for the modeling, such data is eliminated due to its extremely limited representation. The project additionally aims to analyze the characteristics and limitations of the dataset, implement the classification algorithm without relying on machine learning frameworks, evaluate its ability to generalize to untrained ECG recordings, and improve the model using a machine learning framework.

III. BACKGROUND AND THEORETICAL FOUNDATIONS

To understand the basic background and theoretical foundations of the arrhythmia classification problem using ECG analysis, a few concepts that are better to know are addressed in the following sections.

A. Electrocardiogram (ECG)

An electrocardiogram (ECG) is a recording of the electrical activity produced by the heart during each cardiac cycle. A typical heartbeat produces a recognizable waveform consisting primarily of the P wave, QRS complex, and T wave. These components represent different stages of electrical activation and recovery of the heart.

The electrical cycle begins with the P wave, which represents atrial depolarization. During this stage, an electrical signal spreads through the atria, causing them to contract and move blood into the ventricles.

After the P wave, the signal travels toward the ventricles and produces the QRS complex, which represents ventricular depolarization. This electrical activation causes the ventricles to contract and pump blood out of the heart. The QRS complex consists of three possible deflections:

- Q wave: The first negative deflection before the main positive peak. It represents the early stage of ventricular depolarization.
- R wave: The main positive deflection and usually the most prominent peak in an ECG heartbeat. It represents a major portion of ventricular depolarization. Because R peaks are generally easy to identify, the distance between consecutive R peaks, known as the RR interval, is commonly used to analyze heart rate and rhythm.
- S wave: The negative deflection following the R wave. It represents the later stages of ventricular depolarization.

Together, the Q, R, and S waves form the QRS complex. The duration and shape, or morphology, of this complex provide important information about how electrical signals travel through the ventricles. Abnormal heartbeats can

produce QRS complexes with different widths, amplitudes, or shapes, making QRS-related features particularly useful for arrhythmia classification.

Following the QRS complex is the T wave, which represents ventricular repolarization. During this process, the ventricular cells electrically recover and return toward their resting state in preparation for the next heartbeat.

A simplified representation of one ECG cycle can therefore be described as:

P wave → QRS complex → T wave

or physiologically as:

Atrial activation → Ventricular activation → Ventricular recovery

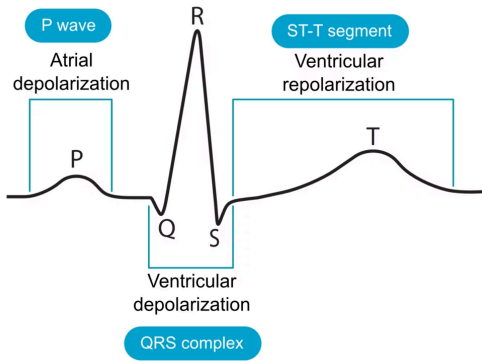


Image 1. Electrocardiogram basic components ^[1]

In addition to the individual waves, ECG analysis considers the intervals between these components. These intervals describe the timing of electrical conduction through different regions of the heart. For example, the RR interval describes the timing between consecutive heartbeats, while the PQ, QRS, QT, and ST measurements provide information about different stages of electrical conduction, ventricular activation, and recovery.

Because arrhythmias can alter the timing, amplitude, duration, and morphology of these ECG components, measurements derived from the P, Q, R, S, and T waves can be used as input features for machine-learning models designed to distinguish normal and abnormal heartbeats.

IV. DATASET AND DATA PREPARATION

A. Dataset Source and Feature Extraction

Before getting into analyzing the dataset, it is important to mention that the ECG features used in this study^[2] were not extracted directly as part of the proposed machine-learning pipeline. Instead, the dataset provides features previously extracted from the original MIT-BIH Arrhythmia Database^[3] according to the feature-extraction methodology presented by Sakib, Fouda, and Fadlullah^[4]. Their methodology consists of ECG preprocessing, heartbeat segmentation, peak detection, and feature extraction. The resulting features are organized into four principal categories: RR interval

features, heartbeat interval features, heartbeat wave-amplitude features, and QRS morphology features. Measurements are extracted from the two ECG channels available in the original recordings, providing two sets of features for each heartbeat. These numerical features are subsequently used as the input variables for the classification models developed in this study.

B. Dataset analysis

The dataset contains 100,689 heartbeat instances and 34 columns, including the ECG-derived features, the heartbeat classification label (type), and the source ECG recording identifier (record). Table 1 summarizes the physiological or computational meaning of the extracted features and Table 2 explains the heartbeat classes (type) and the source identifier(record).

Feature	Meaning	Description / Relevance
pre-RR	Previous RR interval	Time interval between the R peak of the previous heartbeat and the R peak of the current heartbeat. It describes the timing of the current beat relative to the preceding beat and can help identify premature or irregular beats.
post-RR	Following RR interval	Time interval between the R peak of the current heartbeat and the R peak of the following heartbeat. Together with pre-RR, it provides information about local heart rhythm and irregularities.
pPeak	P-wave peak amplitude	Amplitude of the P wave, which represents atrial depolarization. Variations in its amplitude and morphology may indicate changes in atrial electrical activity.
qPeak	Q-wave peak amplitude	Amplitude of the Q wave, the initial negative deflection of the QRS complex. It contributes information about the morphology of ventricular depolarization.
rPeak	R-wave peak amplitude	Amplitude of the R wave, typically the largest positive deflection of the QRS complex. It provides information about ventricular electrical activity and QRS morphology.

sPeak	S-wave peak amplitude	Amplitude of the S wave, the negative deflection following the R wave. It provides additional information about the shape of the QRS complex.
tPeak	T-wave peak amplitude	Amplitude of the T wave, which represents ventricular repolarization. Changes in the T wave can reflect variations in ventricular recovery.
qrs_inter val	QRS duration	Duration of the QRS complex. It represents the time associated with ventricular depolarization and is useful for distinguishing beats with abnormal ventricular conduction or morphology.
pq_inter val	PQ interval	Time interval associated with conduction from atrial depolarization toward the beginning of ventricular depolarization. It provides information about atrioventricular conduction.
qt_inter val	QT interval	Time from the beginning of ventricular depolarization to the completion of ventricular repolarization. It represents the overall duration of ventricular electrical activity.
st_inter val	ST interval	Interval associated with the period between ventricular depolarization and repolarization. It provides information about the transition from the QRS complex to the T wave.
qrs_morph0	QRS morphology value 0	A sampled/derived value representing part of the morphology (shape) of the QRS complex.
qrs_morph1	QRS morphology value 1	A second value used to characterize the shape of the QRS complex.
qrs_morph2	QRS morphology value 2	A third value describing QRS morphology.

qrs_morph3	QRS morphology value 3	A fourth value describing QRS morphology.
qrs_morph4	QRS morphology value 4	A fifth value describing QRS morphology. Together, the morphology features allow the model to distinguish different QRS shapes.

Table 1. Description of each ECG feature

Column	Meaning	Role
record	MIT-BIH ECG record identifier	Identifies the source ECG recording from which the heartbeat was extracted. It should generally be treated as an identifier/grouping variable rather than a predictive physiological feature.
type	Heartbeat class	Target variable indicating the heartbeat/arrhythmia category to be predicted by the classification model.

Table 2. Description of type and record

C. ECG Channel Representation

The dataset contains ECG measurements obtained from two different channels, represented by the prefixes 0_ and 1_. Both channels record the electrical activity of the same heartbeat, but from different ECG signals, providing different representations of the heart's electrical activity. Consequently, the same set of features is extracted from each channel. For example, 0_rPeak represents the R-wave peak measured from Channel 0, while 1_rPeak represents the corresponding measurement from Channel 1. Using two channels can provide complementary information because the amplitude and morphology of an ECG waveform can appear differently depending on the signal from which it is measured. Therefore, the dataset contains two sets of the same ECG features, allowing the classification model to use information from both channels when distinguishing between different heartbeat types. The following Table explains the data type, observed range and the measurement unit for each ECG feature.

Feature	Type	Channel 0 Range	Channel 1 Range	Unit
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pre-RR	Integer	11 – 2,114	11 – 2,114	Extracted interval
post-RR	Float	6 – 2,114	6 – 2,114	Extracted interval
pPeak	Float	−3.094 – 3.821	−2.108 – 5.161	Signal amplitude
qPeak	Float	−3.871 – 3.081	−5.482 – 4.034	Signal amplitude
rPeak	Float	−2.373 – 4.019	−3.847 – 4.034	Signal amplitude
sPeak	Float	−3.623 – 4.019	−6.155 – 3.318	Signal amplitude
tPeak	Float	−2.006 – 4.118	−2.493 – 7.003	Signal amplitude
qrs_inter val	Integer	0 – 170	0 – 260	Extracted interval
pq_inter val	Integer	1 – 150	1 – 185	Extracted interval
qt_inter val	Integer	5 – 334	5 – 412	Extracted interval
st_inter val	Integer	1 – 141	1 – 215	Extracted interval
qrs_morph0	Float	−3.871 – 3.081	−5.482 – 4.034	Signal amplitude
qrs_morph1	Float	−2.629 – 2.870	−4.494 – 3.586	Signal amplitude
qrs_morph2	Float	−2.840 – 3.331	−4.398 – 2.692	Signal amplitude
qrs_morph3	Float	−2.995 – 3.225	−4.070 – 3.756	Signal amplitude
qrs_morph4	Float	−3.086 – 2.633	−4.520 – 2.869	Signal amplitude

Table 3. Data types, observed ranges, and measurement units of the ECG features used in the dataset.

The features in the dataset are represented primarily through extracted intervals and signal-amplitude values. An extracted interval describes the distance between two characteristic points of the ECG waveform. Since an ECG is

a digital signal, it consists of a sequence of individual measurements called samples. The original MIT-BIH Arrhythmia Database was recorded at a sampling frequency of 360 Hz, meaning that 360 signal samples were collected per second. Therefore, distances between detected ECG points can be represented as the number of samples separating them. Features such as the RR, PQ, QT, ST, and QRS intervals belong to this category and provide information about the timing of cardiac electrical activity.

Signal amplitude, on the other hand, represents the magnitude of the ECG signal at a particular point. In an ECG waveform, the horizontal axis represents time or sample position, while the vertical axis represents signal amplitude. Positive and negative values indicate the direction and magnitude of the signal relative to its baseline. The P, Q, R, S, and T peak features use these amplitude values to characterize important parts of the heartbeat waveform. The QRS morphology features similarly contain values derived from the ECG signal to describe the shape of the QRS complex. Since these values were obtained after ECG preprocessing and feature extraction, they are treated as processed signal-amplitude values rather than assumed to represent direct clinical measurements in millivolts.

D. Target Variable and Record Identifier

In addition to the extracted ECG features, the dataset contains two columns with specific roles in the classification pipeline: type and record. The type column represents the target variable and assigns each heartbeat to one of five categories: Normal (N), Supraventricular Ectopic Beat (SVEB), Ventricular Ectopic Beat (VEB), Fusion Beat (F), or Unknown Beat (Q). Since each heartbeat is assigned to a single category, the task is formulated as a mutually exclusive multiclass classification problem.

The record column identifies the source ECG recording from which each heartbeat was extracted. It is not considered a predictive ECG feature and is therefore not provided as an input to the classification model. However, it is retained during data preparation to construct the training and test sets at the recording level, as described in the preprocessing section.

V. EXPLORATORY DATA ANALYSIS

A. Data Quality Assessment

Prior to modeling, the dataset was examined for missing values, duplicated observations, and abnormalities in the numerical features. As a result, it is discovered that there are not any missing or duplicated values. However, there were two different sets of features that contained the exact same values, which were 0_pre-RR with 1_pre-RR and 0_post-RR with 1_post-RR. Based on the observations, two new features called pre-RR and post-RR were created, with the following deletion of the features: 0_pre-RR, 1_pre-RR, 0_post-RR, and 1_post-RR

B. Target Class Distribution

Based on the distribution of the 5 distinct classes, it was found that there is an imbalance of the distribution between

the classes. The Normal(N) heartbeat accounts for 89.4% of the classes, meanwhile Unknown beat(Q) comprises approximately 0.014 %. Ventricular Ectopic Beat (VEB) constitutes 6.9%, Supraventricular Ectopic Beat (SVEB) forms 2.7% , and Fusion Beat (F) makes up 0.79% of the target data. This imbalance can affect the training process because the model is exposed to significantly more examples from the majority class. This can result that the optimization process may favor correctly classifying Normal (N) heartbeats while giving less importance to errors involving minority classes. As a result, the model may achieve high overall accuracy while still performing poorly at detecting less frequent arrhythmia classes. Therefore, evaluation metrics such as precision, recall, F1-score, and macro-F1 are considered in addition to accuracy when evaluating the classification model.

In addition, the Q category was subsequently excluded from the final classification task. It contained only 15 observations out of 100,689 heartbeats, corresponding to approximately 0.015% of the original dataset. Under the record-level partition, these observations could not be reliably represented across training, validation, and test subsets, making model selection and performance estimation for this class unreliable. In addition, generating large numbers of synthetic Q observations from such a small number of original examples would provide limited genuine information.

In contrast, the Fusion Beat (F) category was retained. Although F represented only approximately 0.80% of the original dataset, it contained 803 real observations and represents a defined heartbeat category. Therefore, the final classification problem consisted of N, SVEB, VEB, and F.

C. Feature Distribution Analysis

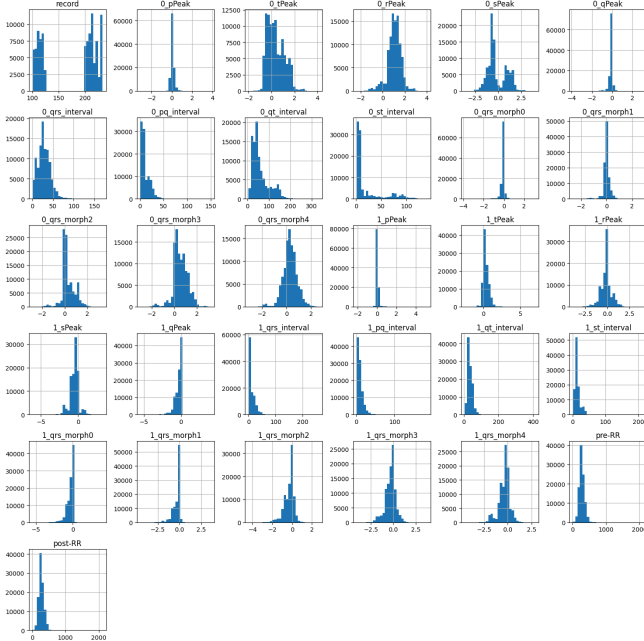


Image 2. Feature distributions

The feature distributions based on the Image 2 show differences in scale, skewness, variability, and distribution shape. Interval-based features generally exhibit positively skewed distributions and wider numerical ranges, whereas amplitude and morphology features are concentrated within smaller ranges. Some features also exhibit multiple regions of concentration, which may reflect differences among heartbeat classes. These observations motivated feature standardization before training scale-sensitive models, while extreme observations were retained because they may represent genuine cardiac variability.

D. Feature Correlation and Redundancy

A correlation analysis was performed to identify relationships and potential redundancy among the numerical features. The correlation matrix showed several strong relationships, particularly among the QRS morphology features. Notably, 0_qPeak and 0_qrs_morph0, as well as 1_qPeak and 1_qrs_morph0, showed perfect positive correlations. Therefore, the two qrs_morph0 features were removed while the corresponding qPeak features were retained due to their clearer physiological interpretation.

VI. DATA PREPROCESSING

A. Record-Level Train-Validation-Test Split

The record variable identifies the ECG recording from which each heartbeat was extracted. Since multiple heartbeat samples can originate from the same recording, randomly splitting individual samples could place heartbeats from the same recording in both the training and test sets. These samples may share recording-specific characteristics, potentially resulting in an overly optimistic evaluation of model performance.

Therefore, records were assigned approximately 60% to training, 20% to validation, and 20% to testing. Because individual ECG recordings contain different numbers of heartbeat observations, this resulted in approximately 58%, 17%, and 25% of heartbeat samples in the training, validation, and test subsets, respectively. This ensures that records used for validation and testing were not observed during training and provides a more rigorous evaluation of the model's ability to generalize to unseen ECG recordings. The record variable was used only for the data split and was excluded from the model's input features.

B. Feature Standardization

Standardization was performed because the ECG features were represented on substantially different numerical scales like we observed during the feature distribution analysis. Each feature was transformed to have approximately zero mean and unit standard deviation. To prevent information leakage, the mean and standard deviation were calculated from the training data. The same training-derived parameters were subsequently used to transform the validation and test data to standardize the transformation of the data.

C. Principal Component Analysis

Principal Component Analysis (PCA) was applied after standardization to investigate whether the dimensionality of the feature space could be reduced while preserving most of its variability. PCA was calculated exclusively from the standardized training data, and the resulting principal component transformation was subsequently applied to the validation and test data.

A cumulative explained variance threshold of 95% was selected. The first 14 principal components explained approximately 94.76% of the total variance, while 15 components explained approximately 96.11%. Therefore, 15 principal components were retained, representing the smallest number of components exceeding the selected 95% threshold.

VII. CLASSIFICATION METHODOLOGY

A. Logistic regression

Logistic regression was selected as the initial model. Logistic regression provides a simple and interpretable approach to classification and gives insights of whether more complex models or preprocessing strategies are needed subsequently. Beginning with a linear classifier also makes it possible to determine how effectively the extracted ECG features separate the heartbeat classes without introducing additional model complexity.

B. Multiclass Output and Softmax

For multinomial logistic regression, the linear output of the model is calculated as:

$$Z = WX + b$$

X is the input matrix that contains all the heartbeat samples used by the model. Each row represents one heartbeat, while each column represents one input feature. After PCA, each heartbeat is represented by 15 principal components.

W is the weight matrix that contains the weights learned by the logistic regression model during training. Since there are four possible heartbeat classes, the model learns a separate set of 15 weights for each class.

b is the bias vector that contains one additional bias value for each of the four classes. These values are also learned during training.

Lastly, Z is the logit matrix that contains the scores calculated by the model. For every heartbeat, the model produces four scores, one for each possible class. These scores are then converted into probabilities using the softmax function.

$$\hat{y}_{ik} = \frac{e^{z_{ik}}}{\sum_{j=1}^K e^{z_{ij}}}$$

Here, Softmax was selected as the output activation function because the ECG classification problem consists of four mutually exclusive classes: N, SVEB, VEB, and F. Each heartbeat is assigned to only one of these classes. Softmax converts the model's output scores into a probability distribution across all four classes, where the

probabilities sum to one. This allows the class with the highest predicted probability to be selected as the final prediction. In contrast, applying sigmoid independently to each output would treat each class as an independent binary classification problem, allowing multiple classes to receive high probabilities simultaneously, which is not appropriate for this mutually exclusive classification task.

C. Cross-Entropy Loss

$$L = -\frac{1}{m} \sum_{i=1}^m \sum_{k=1}^K y_{ik} * \log(\hat{y}_{ik})$$

Categorical cross-entropy was used as the loss function to quantify the difference between the predicted class probabilities and the true one-hot encoded labels. The loss penalizes the model when a low probability is assigned to the correct heartbeat class and provides the optimization objective for training.

D. Gradient Descent

To train the logistic regression model, gradient descent was used to minimize the categorical cross-entropy loss. The objective of this optimization process is to determine the weights and biases that minimize the difference between the predicted class probabilities and the actual heartbeat classes.

$$\frac{\partial L}{\partial z} = \hat{y} - y$$

The gradients represent how much the loss changes with respect to each model parameter. The difference between the predicted probabilities and the true labels is first calculated and then used to obtain the gradients for the weight matrix and bias vector. These gradients indicate both the direction and magnitude in which the parameters should be adjusted to reduce the loss.

$$\frac{\partial L}{\partial W} = \frac{1}{m} X^T (\hat{y} - y)$$

$$\frac{\partial L}{\partial b} = \frac{1}{m} \sum_{i=1}^m (\hat{y} - y)$$

The weights and biases are updated in the opposite direction of their respective gradients because the objective is to minimize the loss function. The learning rate, in this case set as 0.01, controls the size of each update. This process is repeated over multiple epochs, allowing the model parameters to progressively adjust toward values that produce a lower classification loss.

$$W \leftarrow W - \alpha \frac{\partial L}{\partial W}$$

$$b \leftarrow b - \alpha \frac{\partial L}{\partial b}$$

Batch gradient descent was selected for the baseline model because it provides a straightforward implementation of the optimization process. In each epoch, all training observations are used to calculate the gradients before performing a single parameter update. This produces a

stable estimate of the gradient without the fluctuations. Furthermore, the relatively small number of model parameters after PCA made full-batch optimization computationally feasible for this experiment.

Therefore, during each epoch, the complete training dataset was passed through the model, the cross-entropy loss and corresponding gradients were calculated, and the weights and biases were updated once. This procedure was repeated for 1,000 epochs.

VIII. EVALUATION METRICS

To evaluate the performance of the classification model, several metrics were considered. Since the dataset presents a class imbalance, overall accuracy alone may not adequately represent the model's ability to classify the less frequent heartbeat categories. Therefore, accuracy was complemented with precision, recall, F1-score, macro-F1, and a confusion matrix.

A. Accuracy

$$Accuracy = \frac{\text{Correct predictions}}{\text{Total predictions}}$$

Accuracy represents the proportion of heartbeat samples that are correctly classified out of the total number of samples. Although it provides a general measure of model performance, it can be misleading for imbalanced datasets. In this dataset, Normal (N) heartbeats represent the majority of observations; therefore, a model could obtain high accuracy by correctly predicting the N class while performing poorly on the minority arrhythmia classes.

B. Precision

$$Precision = \frac{TP(\text{True positive})}{TP + FP(\text{False Positive})}$$

Precision measures how many of the samples predicted as a particular class actually belong to that class. A high precision indicates that when the model predicts a specific heartbeat category, the prediction is usually correct.

C. Recall

$$Recall = \frac{TP}{TP + FN(\text{False Negative})}$$

Recall measures the proportion of samples belonging to a particular class that the model successfully identifies. This metric is particularly important in an imbalanced classification problem because it reveals whether the model is failing to detect observations from minority classes.

D. F1-Score

$$F1 = 2 \cdot \frac{Precision * Recall}{Precision + Recall}$$

The F1-score combines precision and recall into a single metric using their harmonic mean. A high F1-score

therefore requires both high precision and high recall. This makes the F1-score useful for evaluating each heartbeat class when the class distribution is imbalanced.

E. Macro-F1

$$Macro\ F1 = \frac{1}{K} \sum_{k=1}^K F1_k$$

Macro-F1 is calculated by obtaining the F1-score of each class independently and then calculating their average. Unlike accuracy, each heartbeat category contributes equally to macro-F1 regardless of the number of observations belonging to that class. For this reason, macro-F1 is particularly useful in this study for determining whether the model performs consistently across both majority and minority classes.

F. Confusion Matrix

A confusion matrix was also used to examine the specific classification errors made by the model. Each row represents the actual heartbeat class, while each column represents the class predicted by the model. Values along the main diagonal correspond to correct classifications, whereas values outside the diagonal represent misclassifications. This allows the model's behavior across individual heartbeat categories to be analyzed in greater detail than with a single performance metric.

IX. BASELINE MODEL RESULTS

A. Initial Baseline Configuration

The baseline model consisted of a multinomial logistic regression classifier implemented without the use of machine-learning frameworks. After standardization and Principal Component Analysis, each heartbeat was represented by 15 principal components. The model was trained using batch gradient descent with a learning rate of 0.01 for 1,000 epochs.

The performance of the initial model was evaluated on both the training and validation subsets. The training set was used to estimate the parameters of the model, while the validation set was used to evaluate its ability to generalize to ECG recordings that were not observed during training and to guide the selection of the baseline hyperparameters.

Dataset	Accuracy	Macro-F1
Training	0.9075	0.4157
Validation	0.9460	0.3630

Table 4. Initial Baseline Training and Validation Performance

Although the validation accuracy of 94.60% appears high, the validation macro-F1 score was only 0.3630. This difference indicates that accuracy alone does not adequately represent the performance of the model because of the severe class imbalance in the dataset. The model correctly classified a large proportion of the majority Normal (N) observations while showing substantially weaker performance on the minority heartbeat categories.

B. Baseline Hyperparameter Experiment

To investigate whether a smaller learning rate and a greater number of training epochs could improve the generalization of the baseline classifier, a second configuration was evaluated. The learning rate was reduced from 0.01 to 0.001, while the number of epochs was increased from 1,000 to 2,000. All other preprocessing and modeling procedures were maintained.

Configuration	Learning Rate	Epochs	Validation Accuracy	Validation Macro-F1
Baseline 1	0.01	1,000	0.9460	0.3630
Baseline 2	0.001	2,000	0.9169	0.3478

Table 5. Baseline Hyperparameter Comparison

The second configuration did not improve the performance of the model. Its validation accuracy decreased from 94.60% to 91.69%, while macro-F1 decreased from 0.3630 to 0.3478. Therefore, increasing the number of epochs while decreasing the learning rate did not produce better generalization.

Because macro-F1 gives equal importance to all heartbeat categories and is more appropriate for the highly imbalanced target distribution, it was used as the principal metric for model selection. Consequently, the first configuration, with a learning rate of 0.01 and 1,000 epochs, was retained as the final baseline model.

C. Final Baseline Test Performance

After selecting the baseline configuration using the validation set, the selected model was evaluated on the held-out test set. The test data were not used for hyperparameter selection and were reserved for evaluating the final generalization performance of the selected baseline model.

Dataset	Accuracy	Macro-F1
Training	0.9075	0.4157

Validation	0.9460	0.3630
Test	0.9587	0.3909

Table 6. Selected Baseline Performance

The selected baseline achieved a test accuracy of 95.87% but a macro-F1 score of only 0.3909. The high accuracy therefore continued to contrast with the considerably lower macro-F1 score, suggesting that the model's overall performance was strongly influenced by the majority N class.

D. Baseline Confusion Matrix and Per-Class Performance

Class	Precision	Recall	F1-score
N	0.9584	0.9995	0.9785
SVEB	0.0000	0.0000	0.0000
VEB	0.9720	0.4184	0.5850
F	0.0000	0.0000	0.0000

Table 7. Baseline Test Per-Class Performance

The class-specific results demonstrate that the high test accuracy was primarily produced by strong performance on the N class. The model obtained an N F1-score of 0.9785 and correctly identified nearly all Normal observations. VEB also achieved a high precision of 0.9720. However, its recall of 0.4184 indicates that less than half of the actual VEB observations were correctly detected.

In contrast, the model obtained F1-scores of zero for SVEB and F, indicating that it failed to correctly identify these minority classes. These results demonstrate the limitations of the linear baseline model when applied to the highly imbalanced heartbeat classification problem.

E. Baseline Model Assessment

Despite achieving a test accuracy of 95.87%, the baseline model obtained a test macro-F1 of only 0.3909. This discrepancy demonstrates the effect of class imbalance on the classifier. Because most observations belong to the N class, strong performance on this class produces high overall accuracy even when several minority heartbeat categories are poorly classified.

To evaluate the bias, variance, and level of fit of the model, training and validation performance were compared. The baseline obtained a training macro-F1 of 0.4157 and a validation macro-F1 of 0.3630, producing a relatively small generalization gap of 0.0527. The similarly low macro-F1

values indicate that the model performs poorly across the four classes even on the training data. Therefore, the baseline model is considered to have high bias. In contrast, because the difference between training and validation macro-F1 is relatively small, the model is considered to have low variance.

Based on this combination of high bias and low variance, the baseline logistic regression is classified as underfitting. The model does not appear to be excessively adapting to the training data; instead, its linear structure and the severe class imbalance limit its ability to represent the patterns required to distinguish the minority heartbeat classes.

Model	Bias	Variance	Level of Fit
Baseline Logistic Regression	High	Low	Underfitting

Table 8. Baseline Model’s Bias, Variance and Level of Fit.

X. XGBOOST MODEL IMPROVEMENT

A. Motivation for XGBoost

The baseline multinomial logistic regression achieved high overall accuracy but relatively low macro-F1, demonstrating limited performance across the minority heartbeat classes. In particular, the linear baseline was unable to correctly identify several of the less frequent heartbeat categories. Therefore, a more flexible classification approach was investigated to improve the model’s ability to distinguish the four heartbeat classes.

XGBoost was selected as the improved model because, unlike logistic regression, it can capture nonlinear relationships and interactions among the ECG features. This is particularly relevant for the present dataset, where heartbeat classification may depend on combinations of rhythm, interval, amplitude, and QRS morphology characteristics rather than on simple linear relationships between individual features and the target class.

XGBoost is an ensemble learning method based on gradient-boosted decision trees. Instead of constructing a single decision tree, the algorithm builds multiple trees sequentially. At each boosting iteration, a new tree is constructed to improve the errors of the current ensemble. The contribution of each new tree is controlled by a learning rate, allowing the model to progressively improve its predictions while limiting the influence of each individual tree.

For the XGBoost model, the original ECG features after the feature-selection and standardization steps were used

instead of the 15 principal components used by the baseline logistic regression. PCA was not applied because the tree-based structure of XGBoost can directly model nonlinear relationships and interactions among the original ECG features. Although feature standardization is not required for decision-tree models, the training-derived standardized features from the existing preprocessing pipeline were retained for consistency.

B. Addressing Class Imbalance Through Random Oversampling

The severe class imbalance identified during exploratory data analysis remained an important limitation for model training. Normal (N) heartbeats represented the large majority of the observations, minority classes such as SVEB and particularly F contained substantially fewer examples than N. To increase the influence of the minority heartbeat categories during XGBoost training, random oversampling was applied exclusively to the training subset.

Random oversampling increases the representation of minority classes by randomly duplicating their existing observations. The procedure was applied only after the record-level train-validation-test split and only to the training data. The validation and test subsets were left unchanged so that model performance could be evaluated using their original class distributions.

Although oversampling allows the classifier to receive greater exposure to minority-class observations, it does not create genuinely new ECG patterns. Consequently, repeatedly presenting the same rare observations to a high-capacity model may increase the risk of memorization and overfitting. This behavior was investigated by comparing the model’s training and validation performance.

C. XGBoost Configuration

The initial oversampled XGBoost classifier was configured with 300 boosting estimators, a maximum tree depth of 6, and a learning rate of 0.1. A subsample ratio of 0.8 was used so that each boosting iteration used a random subset of the training observations, while column sample by tree with 0.8 restricted each tree to a subset of the available features. These mechanisms introduce randomness into the boosting process and can help reduce model variance.

The multiclass probabilistic objective was used for the four heartbeat categories, and multiclass logarithmic loss was used as the internal evaluation objective.

Hyperparameter	Value
n_estimators	300
max_depth	6
learning_rate	0.1

subsample	0.8
colsample_bytree	0.8
reg_lambda	1
reg_alpha	0
gamma	0
min_child_weight	1

Table 9. Initial hyperparameters of XGBoost

XI. XGBOOST RESULTS AND REGULARIZATION

A. Initial Oversampled XGBoost Performance

The initial oversampled XGBoost model achieved near-perfect performance on the training subset, with an accuracy of 1.0000 and macro-F1 of 0.9999. Validation accuracy was 0.9618 and validation macro-F1 was 0.4567.

Dataset	Accuracy	Macro Precision	Macro Recall	Macro-F1
Training	1.0000	0.9999	1.0000	0.9999
Validation	0.9618	0.4443	0.4783	0.4567

Table 10. Initial Oversampled XGBoost Performance

Random oversampling substantially increased training performance and produced a validation macro-F1 of 0.4567. However, the training macro-F1 of 0.9999 resulted in a generalization gap of approximately 0.5432. The near-perfect training performance combined with substantially lower validation performance indicates low bias but high variance and therefore strong overfitting.

B. Oversampled XGBoost Regularization Experiment

Because the initial XGBoost model exhibited substantial overfitting, stronger regularization was evaluated. The experiments modified tree depth, L2 regularization, L1 regularization, minimum split gain, and minimum child weight. These parameters constrain either the magnitude of leaf predictions or the complexity of the individual decision trees.

Index	Depth	λ	α	γ	Min Child Weight
1	6	1	0.0	0.0	1

2	5	5	0.0	0.0	2
3	4	5	0.1	0.1	3
4	4	10	0.5	0.2	5
5	3	10	1.0	0.5	5

Index	Train Macro-F1	Val Macro-F1	Gap
1	0.9999	0.4567	0.5432
2	0.9936	0.4526	0.5409
3	0.9774	0.4516	0.5258
4	0.9755	0.4583	0.5172
5	0.9419	0.4449	0.4970

Table 11. Result of the XGBoost Regularization Experiment

Stronger regularization generally reduced training macro-F1 and decreased the training-validation gap. The fourth configuration produced the highest validation macro-F1 among the random-oversampling configurations at 0.4583. However, the improvement over the initial value of 0.4567 was small, and a substantial generalization gap remained.

C. Synthetic Minority Oversampling Technique (SMOTE)

Since random oversampling only duplicates existing minority-class observations, an additional experiment was performed using the Synthetic Minority Oversampling Technique (SMOTE). Instead of directly replicating existing samples, SMOTE generates synthetic minority observations by interpolating between neighboring samples in the feature space.

SMOTE was applied only to the training subset after the record-level split and feature standardization. Validation and test observations were left unchanged. This prevents synthetic training information from entering the evaluation subsets and preserves an independent estimate of model generalization.

The training distribution before SMOTE contained 50,456 N, 2,344 SVEB, 4,789 VEB, and 759 F observations. SMOTE was configured to balance each minority class to the size of the majority class, resulting in 50,456 observations per class in the resampled training data. The parameter $k_neighbors$ was set to 1 in the experiment.

D. Initial SMOTE XGBoost Performance

Dataset	Accuracy	Macro Precision	Macro Recall	Macro-F1
Training	1.0000	0.9999	1.0000	0.9999
Validation	0.9686	0.4674	0.4942	0.4754

Table 12. Initial SMOTE XGBoost performance

The initial SMOTE-XGBoost configuration achieved nearly perfect performance on the original training subset, with a macro-F1 of 0.9999. Validation macro-F1 reached 0.4754, exceeding both the baseline logistic regression and the random-oversampling XGBoost configurations.

Nevertheless, the difference between training and validation macro-F1 remained substantial as 0.5245. This large generalization gap indicates that the SMOTE-XGBoost model has low bias but high variance, and is therefore classified as overfitting.

Class	Precision	Recall	F1
N	0.9907	0.9813	0.9860
SVEB	0.0533	0.1452	0.0779
VEB	0.8257	0.8503	0.8378
F	0.0000	0.0000	0.0000

Table 13. Result of the validation per class

SMOTE notably improved VEB performance and allowed the model to detect some SVEB observations that the baseline failed to identify. However, F remained undetected in validation, with precision, recall, and F1 equal to zero. Therefore, SMOTE improved balanced performance overall but did not solve the difficulty associated with the rarest retained heartbeat category.

E. SMOTE XGBoost Regularization Experiment

For the regularization experiment, the same configuration for the oversampled XGBoost regularization experiment was used.

Index	Depth	λ	α	γ	Min Child Weight
1	6	1	0.0	0.0	1
2	5	5	0.0	0.0	2

3	4	5	0.1	0.1	3
4	4	10	0.5	0.2	5
5	3	10	1.0	0.5	5

Index	Train Macro-F1	Val Macro-F1	Gap
1	0.9999	0.4754	0.5245
2	0.9937	0.4731	0.5206
3	0.9805	0.4740	0.5065
4	0.9791	0.4708	0.5083
5	0.9515	0.4634	0.4881

Table 14. Result of the SMOTE XGBoost Regularization Experiment

Stronger regularization reduced training performance and, in some configurations, decreased the training-validation gap. However, none of the stronger regularization configurations exceeded the validation macro-F1 of 0.4754 achieved by the initial SMOTE XGBoost configuration. Therefore, the initial configuration was retained for final testing.

F. Final XGBoost Test Performance

After selecting the initial SMOTE-XGBoost configuration based exclusively on validation macro-F1, the model was evaluated on the held-out test subset.

Dataset	Accuracy	Macro Precision	Macro Recall	Macro-F1
Training	1.0000	0.9999	1.0000	0.9999
Validation	0.9686	0.4674	0.4942	0.4754
Test	0.9219	0.5009	0.4983	0.4831

Table 15. Final SMOTE XGBoost Performance

It achieved a test accuracy of 92.19% and a macro-F1 of 0.4831. Although its accuracy was lower than the baseline logistic regression, its macro-F1 was substantially higher, indicating better balanced performance across the four heartbeat categories.

Model	Bias	Variance	Level of Fit
SMOTE XGBoost	Low	High	Overfitting

Table 16. Final SMOTE XGBoost Bias, Variance, and Level of fit

In addition, SMOTE XGBoost achieved almost perfect training performance but substantially lower validation performance. This indicates low bias and high variance and therefore overfitting.

Class	Precision	Recall	F1
N	0.9785	0.9391	0.9584
SVEB	0.0643	0.2574	0.1029
VEB	0.9609	0.7968	0.8712
F	0.0000	0.0000	0.0000

Table 17. Result of the test per class

The final SMOTE-XGBoost model maintained strong performance on N and substantially improved the detection of VEB. VEB achieved a test F1-score of 0.8712 compared with 0.5850 for the baseline. In addition, the model was able to identify some SVEB observations, producing a recall of 0.2574 and an F1-score of 0.1029, whereas the baseline obtained an SVEB F1-score of zero.

However, the model continued to fail to identify F observations, resulting in an F1-score of zero for this category. Therefore, although SMOTE improved the overall macro-F1 and the classification of VEB and SVEB, it did not completely resolve the extreme class imbalance. The F category remains the principal limitation of the final classifier.

XII. FINAL MODEL COMPARISON

Model	Validation Accuracy	Validation Macro-F1	Test Accuracy	Test Macro-F1
Baseline Logistic Regression	0.9460	0.3630	0.9587	0.3909

Random-Oversampling XGBoost	0.9618	0.4567	-	-
SMOTE XGBoost	0.9686	0.4754	0.9219	0.4831

Table 18. Model Performance Comparison for the test

The comparison demonstrates why accuracy alone is insufficient for evaluating this highly imbalanced classification problem. The baseline logistic regression achieved the highest test accuracy at 95.87%, but its test macro-F1 was only 0.3909. Random-oversampling XGBoost improved validation macro-F1 to 0.4567, while SMOTE-XGBoost achieved the highest validation macro-F1 of 0.4754 and was therefore selected for final testing.

The selected SMOTE-XGBoost obtained a test macro-F1 of 0.4831, representing an increase of approximately 0.0922 over the baseline's 0.3909. This improvement occurred despite a reduction in overall accuracy from 95.87% to 92.19%, demonstrating that the improved model sacrificed some majority-class accuracy in exchange for stronger performance across minority categories.

Nevertheless, the final model exhibited a large training-validation gap and failed to identify F observations. Therefore, the improvement should be interpreted as better balanced classification rather than a complete solution to the class-imbalance problem.

XIII. CONCLUSION

This study developed and evaluated machine-learning approaches for classifying ECG heartbeats using extracted features from the MIT-BIH Arrhythmia Database. Although the original dataset contained five heartbeat labels, the Unknown Beat (Q) category was excluded from the final classification problem because it contained only 15 observations and could not be reliably represented and evaluated under the record-level train-validation-test partition. The Fusion Beat (F) category was retained because it contained 803 real observations and represents a defined heartbeat category, despite comprising only approximately 0.80% of the original dataset.

A four-class multinomial logistic regression model implemented without machine-learning frameworks was first established as the baseline. The selected baseline achieved a test accuracy of 95.87% and a test macro-F1 of 0.3909. Comparison of its training and validation macro-F1 scores indicated high bias, low variance, and underfitting. In particular, the baseline failed to identify SVEB and F observations in the test subset.

To improve minority-class classification, XGBoost was evaluated together with random oversampling and SMOTE. Random oversampling increased validation macro-F1 to

0.4567, while the initial SMOTE XGBoost configuration achieved the highest validation macro-F1 of 0.4754 and was selected for final testing. Stronger regularization reduced model capacity but did not improve upon the selected SMOTE configuration.

The final SMOTE XGBoost model achieved a test accuracy of 92.19% and a macro-F1 of 0.4831. Compared with the baseline, the model substantially improved VEB classification and was able to identify some SVEB observations. However, it continued to fail to identify F heartbeats. In addition, the near-perfect training macro-F1 of 0.9999 compared with a validation macro-F1 of 0.4754 indicates low bias, high variance, and substantial overfitting.

Overall, the experiments demonstrate that high overall accuracy does not necessarily represent effective multiclass classification when the target distribution is severely imbalanced. The lower accuracy but higher macro-F1 of the final SMOTE-XGBoost model shows that addressing imbalance and introducing nonlinear modeling can improve balanced performance across heartbeat classes. At the same time, the persistent difficulty with F and the large training-validation gap demonstrate that SMOTE does not completely solve the limitations created by rare classes and limited recording diversity. Future work could investigate alternative imbalance-handling techniques, additional ECG recordings containing rare heartbeat categories, and model-selection strategies designed to improve minority-class generalization.

References

- [1] ECG Waves. (2023, June). What does the ECG show: P, QRS, ST, T waves (Atrial and ventricular depolarization and repolarization) [Diagram]. ECG Waves. <https://ecgwaves.com/wp-content/uploads/2023/06/what-does-the-ecg-show-p-qrs-st-t-atrial-ventricular-depolarization-repolarization.webp>
- [2] Sakib, S. (2021, August 20). *ECG Arrhythmia Classification Dataset*. Kaggle. <https://www.kaggle.com/datasets/sadmansakib7/ecg-arrhythmia-classification-dataset>
- [3] Moody, G. B., & Mark, R. G. (2001). The impact of the MIT-BIH Arrhythmia Database. *IEEE Engineering in Medicine and Biology Magazine*, 20(3), 45–50. DOI: 10.1109/51.932724.
- [4] Sakib, S., Fouda, M. M., & Fadlullah, Z. M. (2021). Harnessing Artificial Intelligence for Secure ECG Analytics at the Edge for Cardiac Arrhythmia Classification. In *Secure Edge Computing*, pp. 137–153. DOI: 10.1201/9781003028635-11.